

Using a Calculator

It does exactly what you type: brackets round every top and bottom, the right key, the full display, then round.



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THE TOPIC ON ONE MAP

Using a Calculator

01 Fractions, brackets and the full display

$$\frac{38.5 \times 14.2}{18.4 - 5.9}$$

“all the figures on your display”

then to 1 s.f.

02 Powers, roots and standard form

$$\sqrt[3]{340}$$

$$256^{0.25}$$

$$\sqrt[5]{3.2 \times 10^{11}}$$

$\times 10^x$ key

03 Trigonometry on the calculator

$$\sqrt{\frac{\tan 60^\circ + 1}{\tan 60^\circ - 1}}$$

degree mode

04 Check it is sensible

approximations

within 0.1 of 3.14?

KEY MAP The keys that cost marks when they are missing:

ZONE ()

Brackets

round every top,
every bottom
and everything
under a root

ZONE ■ / □

Fraction key

types the
fraction line for
you

ZONE $x^{\square}$

Power key

$1.3^{2.5}$, $32^{0.4}$

ZONE $\times 10^x$

Standard form

2.8×10^{-3} in one
go

01 Fractions, Brackets and the Full Display

Use the fraction key, or put brackets around the whole top and the whole bottom. Write **every** figure first — that is the method mark — then round.

WHEN YOU SEE → YOU DO

“write down all the figures on your calculator display”

→ copy the whole display, no rounding

“give your answer as a decimal”

→ press $S \leftrightarrow D$

“to 1 s.f. / 4 d.p.” in part (b)

→ round your part (a) answer

a long fraction bar

→ $(\text{top}) \div (\text{bottom})$

Fraction key

Type top then bottom

Write full display

Then round

WORKED MODEL • FULL DISPLAY

Work out $\frac{38.5 \times 14.2}{18.4 - 5.9}$. (b) Give it to 1 s.f.

1 $(38.5 \times 14.2) \div (18.4 - 5.9)$

brackets

2 43.736

full display

ANSWER (b) 40

× **TRAP** $38.5 \times 14.2 \div 18.4 - 5.9$ without brackets gives 23.8 — the calculator subtracts 5.9 only at the end.

IDEA 01

Practise: Fractions, Brackets and the Full Display

My questions first, then selected practice.

Q1 **DR ESLAM** Full display

CH 1 • TOPIC 11 2 marks

Work out $\frac{12.6 + 3.9}{2.5 \times 0.8}$. Write down all the figures on your display.

SELECTED PRACTICE**Q2** **PRACTICE** Practice • Easy

CH 1 • TOPIC 11 3 marks

1 (a) Use your calculator to work out $\frac{38.5 \times 14.2}{18.4 - 5.9}$

Write down all the figures on your calculator display.
You must give your answer as a decimal.

(2 marks)

(b) Write your answer to part (a) correct to 1 significant figure.

(1 mark)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 11 3 marks

3 (a) Use your calculator to work out $\frac{\sqrt{7056}}{0.35 \times 12.8}$

Write down all the figures on your calculator display.
You must give your answer as a decimal.

(2 marks)

(b) Write your answer to part (a) correct to 1 significant figure.

(1 mark)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 11 2 marks

7 Work out $\frac{\sqrt{17 + 4^2}}{7.3^2}$

Write down all the figures on your calculator display.

(2 marks)

Q5**PRACTICE**

Practice • Easy

CH 1 • TOPIC 11 1 mark

11 Calculate.

$$\frac{5.39 - 0.98}{0.743 - 0.0743}$$

(1 mark)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 11 3 marks

2 (a) Use your calculator to work out $\frac{\sqrt{70.25}}{4.2 - 2.37}$

Write down all the figures on your calculator display
You must give your answer as a decimal.

(2 marks)

(b) Write your answer to part (a) correct to 4 decimal places.

(1 mark)

Q7**PRACTICE**

Practice • Medium

CH 1 • TOPIC 11 2 marks**10** Calculate.

$$\frac{16.379 - 0.879}{4.2} \times 1.241$$

Give your answer correct to 2 significant figures.

(2 marks)

02 Powers, Roots and Standard Form

Use the power key for any power, the $\sqrt{\quad}$ key for any root, and the $\times 10^x$ key for standard form. Everything under a root goes inside the root.

WHEN YOU SEE → YOU DO

$$\sqrt[3]{340}, \sqrt[5]{3.2 \times 10^{11}}$$

→ $\sqrt{\quad}$ key (SHIFT)

$$1.3^{2.5}, 256^{0.25}$$

→ power key $x^{\square}$

$$10^{\frac{3}{4}}$$

→ power key with a fraction: $10^{(3 \div 4)}$

“give your answer in standard form”

→ convert the display: $4730 = 4.73 \times 10^3$

WORKED MODEL • STANDARD FORM

Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$.

1 top = 273 000

use $\times 10^x$

2 273 000 $\div$ 0.06 = 4 550 000

divide

ANSWER 4.55×10^6

IDEA 02

Practise: Powers, Roots and Standard Form

My questions first, then selected practice.

Q8

DR ESLAM

Root key

CH 1 • TOPIC 11

1 mark

Work out $\sqrt[3]{2744}$.**SELECTED PRACTICE****Q9**

PRACTICE

Practice • Easy

CH 1 • TOPIC 11

2 marks

4 Work out $(5.6 \times 10^6) + (2.3 \times 10^5)$

(2 marks)

Q10**PRACTICE**

Practice · Easy

CH 1 · TOPIC 11 1 mark

10 Calculate.

$$\frac{5}{8} + \sqrt[3]{340}$$

(1 mark)

Q11**PRACTICE**

Practice · Medium

CH 1 · TOPIC 11 2 marks

6 Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

(2 marks)

Q12**PRACTICE**

Practice · Medium

CH 1 · TOPIC 11 1 mark

11 Calculate $\sqrt{256^{0.25} + 4 \times 8}$.

(1 mark)

Q13**PRACTICE**

Practice • Medium

CH 1 • TOPIC 11 1 mark

$$13 \quad \sqrt[3]{2^3 + 2}$$

(1 mark)

Q14**PRACTICE**

Practice • Hard

CH 1 • TOPIC 11 3 marks

5 (a) Use your calculator to work out $\frac{29^2 - 4.6}{\sqrt{35 - 1.9^3}}$

Write down all the figures on your calculator display.

(2 marks)

(b) Write your answer to part (a) correct to 4 significant figures.

(1 mark)

03 Trigonometry on the Calculator

Check the calculator is in **degrees** (a small D on the screen). Close every bracket after sin, cos, tan.

WHEN YOU SEE → YOU DO

sin, cos, tan **in the question**

→ Degree mode first

$\tan 60^\circ + 1$

→ $\tan(60) + 1$ — close the bracket before +1

$(\sin 33^\circ)^2$

→ $(\sin(33))^2$

WORKED MODEL • TRIG INSIDE A ROOT

Work out $\sqrt{\frac{\tan 60^\circ + 1}{\tan 60^\circ - 1}}$.

1 $\sqrt{}$ then fraction: $(\tan(60) + 1) / (\tan(60) - 1)$

brackets

ANSWER 1.931851653

× **TRAP** In radian mode $\tan 60$ is 0.32 — every answer is wrong. Look for the D.

IDEA 03

Practise: Trigonometry on the Calculator

My questions first, then selected practice.

Q15 DR ESLAM Degrees

CH 1 • TOPIC 11 2 marks

Work out $\frac{5 \sin 30^\circ}{\cos 60^\circ}$.

SELECTED PRACTICE**Q16** PRACTICE Practice • Medium

CH 1 • TOPIC 11 2 marks

- 1 Calculate the value of $\sqrt{\frac{\tan 60^\circ + 1}{\tan 60^\circ - 1}}$

Write down all the figures on your calculator display.

You must give your answer as a decimal.

(2 marks)

Q17**PRACTICE**

Practice · Medium

CH 1 · TOPIC 11 3 marks

4 (a) Use your calculator to work out $\sqrt{\frac{\sin 25^\circ + \sin 40^\circ}{\cos 25^\circ - \cos 40^\circ}}$

Write down all the figures on your calculator display.

(2 marks)

(b) Write your answer to part (a) correct to 2 decimal places.

(1 mark)

Q18**PRACTICE**

Practice · Hard

CH 1 · TOPIC 11 3 marks

1 (a) Find the reciprocal of 2.5

(1 mark)

(b) Work out $\sqrt[3]{\frac{4.3 \times \tan 39^\circ}{23.4 - 6.06}}$

Give your answer correct to 3 significant figures.

(2 marks)

Q19**PRACTICE**

Practice • Hard

CH 1 • TOPIC 11 2 marks

3 Work out $\sqrt{\frac{2.5 \times \sin 43^\circ}{8.2^2 - 50.5}}$

Give your answer correct to 3 significant figures.

(2 marks)

Q20**PRACTICE**

Practice • Hard

CH 1 • TOPIC 11 1 mark

6 Use your calculator to work out $\sqrt{1 - (\sin 33^\circ)^2}$.

(1 mark)

04 Check It Is Sensible

Round each number to 1 significant figure (or a nice square), work it out in your head, then compare with the calculator answer.

WHEN YOU SEE → YOU DO

“use approximations to check”

→ each number to 1 s.f., show the working

“is within 0.1 of 3.14”

→ calculate, then $3.04 < \text{value} < 3.24$

WORKED MODEL • APPROXIMATION

$19.42^2 - \sqrt[3]{1006} \div 4.95 = 375.11 \dots$ Check it is sensible.

1 $20^2 - \sqrt[3]{1000} \div 5 = 400 - 2$

approximate

ANSWER $398 \approx 375$: sensible

IDEA 04

Practise: Check It Is Sensible

My questions first, then selected practice.

Q21 DR ESLAM Estimate

CH 1 • TOPIC 11 2 marks

Estimate $\frac{41.2 \times 9.8}{0.49}$.

SELECTED PRACTICE**Q22** PRACTICE Practice • Medium

CH 1 • TOPIC 11 2 marks

7 An approximation for the value of π is given by

$$4\left(1 - \frac{22}{57} + \frac{22}{85} - \frac{22}{105} + \frac{22}{117} - \frac{22}{242}\right)$$

Use your calculator to show that this approximation is within 0.1 of 3.14

(2 marks)

Q23**PRACTICE**

Practice • Hard

CH 1 • TOPIC 11 3 marks

- 4 (a) Use your calculator to work out $19.42^2 - \sqrt[3]{1006} \div 4.95$

Write down your full calculator display.

(1 mark)

- (b) Use approximations to check that your answer to part (a) is sensible.

You **must** show your working.

(2 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | Full display | Fraction key or brackets; write every figure, then round. |
| 02 | Powers / roots | $x^{\square}$, $\sqrt[\square]{}$, $\times 10^x$; convert to standard form. |
| 03 | Trig | Degree mode; close every bracket. |
| 04 | Check | 1 s.f. estimate, compare. |

BOTTOM LINE Brackets → right key → full display → then round → estimate to check.

Answers

Using a Calculator • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • FRACTIONS, BRACKETS AND THE FULL DISPLAY

Q1	Dr Eslam • Full display	8.25
Q2	Practice • Easy	(a) 43.736 (b) 40
Q3	Practice • Easy	(a) 18.75 (b) 20
Q4	Practice • Easy	0.1077981356
Q5	Practice • Easy	6.594885599
Q6	Practice • Medium	(a) 4.580069567 (b) 4.5801
Q7	Practice • Medium	4.6

IDEA 02 • POWERS, ROOTS AND STANDARD FORM

Q8	Dr Eslam • Root key	14
Q9	Practice • Easy	5.83×10^6
Q10	Practice • Easy	7.604532047
Q11	Practice • Medium	4.55×10^6
Q12	Practice • Medium	6
Q13	Practice • Medium	2.15443469
Q14	Practice • Hard	(a) 157.668255 (b) 157.7

IDEA 03 • TRIGONOMETRY ON THE CALCULATOR

- Q15** Dr Eslam • Degrees $\frac{5 \times 0.5}{0.5} = 5$
- Q16** Practice • Medium 1.931851653
- Q17** Practice • Medium (a) 2.75603957 (b) 2.76
- Q18** Practice • Hard (a) 0.4 (b) 0.586
- Q19** Practice • Hard 0.319
- Q20** Practice • Hard 0.8386705679

IDEA 04 • CHECK IT IS SENSIBLE

- Q21** Dr Eslam • Estimate $\frac{40 \times 10}{0.5} = 800$
- Q22** Practice • Medium 3.0418..., and $3.04 < 3.0418 < 3.24$: within 0.1
- Q23** Practice • Hard (a) 375.1121656 (b) $20^2 - 10 \div 5 = 398$: sensible